

Sustainability & Climate Change

Why the future of the planet will be coded, not just conserved

Why should a computer science student sit through a lecture on sustainability?

Because climate technology is quietly becoming one of the fastest-growing, most talent-starved segments of the entire tech industry — and almost nobody in this room is positioned for it yet.

That gap is your opening.

3×

Faster hiring growth in climate-tech roles vs. traditional software roles (illustrative trend)

<2%

Of engineering graduates currently target climate-tech careers — meaning far less competition for you

Your CSE syllabus is already a climate-tech syllabus

Every core subject you have already studied — or are studying right now — has a direct climate application.



Data Structures & Algorithms

Optimizing energy routing and real-time grid load balancing



Machine Learning

Climate forecasting, weather prediction, emissions modeling



Database Systems

ESG reporting and enterprise carbon-accounting platforms



Computer Networks / IoT

Smart grids, precision agriculture, environmental sensor networks



Cloud Computing

Planetary-scale climate simulation and renewable-asset coordination



Cybersecurity

Protecting power grids and critical climate infrastructure

You are not **in the wrong major.**

For fifteen years, sustainability has been taught as an environmental science elective — soil, carbon cycles, ecosystems. That story is now incomplete.

The next chapter of climate action is being written in code — in models, sensors, algorithms and systems. That is your language.

The old assumption

*"Sustainability = Environmental Science."
A subject for biologists, not engineers.*



The new reality

AI, IoT, robotics and software now decide whether climate solutions actually scale.

What we will cover



The climate emergency, in engineering terms

01

Why the numbers are a systems problem, not just a science one.



A trillion-dollar computing problem

02

The scale of capital now chasing climate technology.



Nine technologies rewriting climate action

03

AI, data, IoT, digital twins, robotics, cloud, security, blockchain, software.



Real companies, real code, real impact

04

Case studies you can study, intern at, or compete with.



Careers & research frontiers for you

05

Where CSE graduates are building the next decade of climate work.



Your first move

06

One concrete step to take before you leave this room.

This is a systems failure at planetary scale

+1.5°C

Global warming vs. pre-industrial levels, already breached in recent years

420+ ppm

Atmospheric CO₂ concentration — highest in over 800,000 years

37 Gt

CO₂ emitted globally every year and still rising

\$38 Tn

Projected annual economic damage from climate change by 2050

Every one of these numbers is generated, tracked, modeled, predicted and acted upon by software.

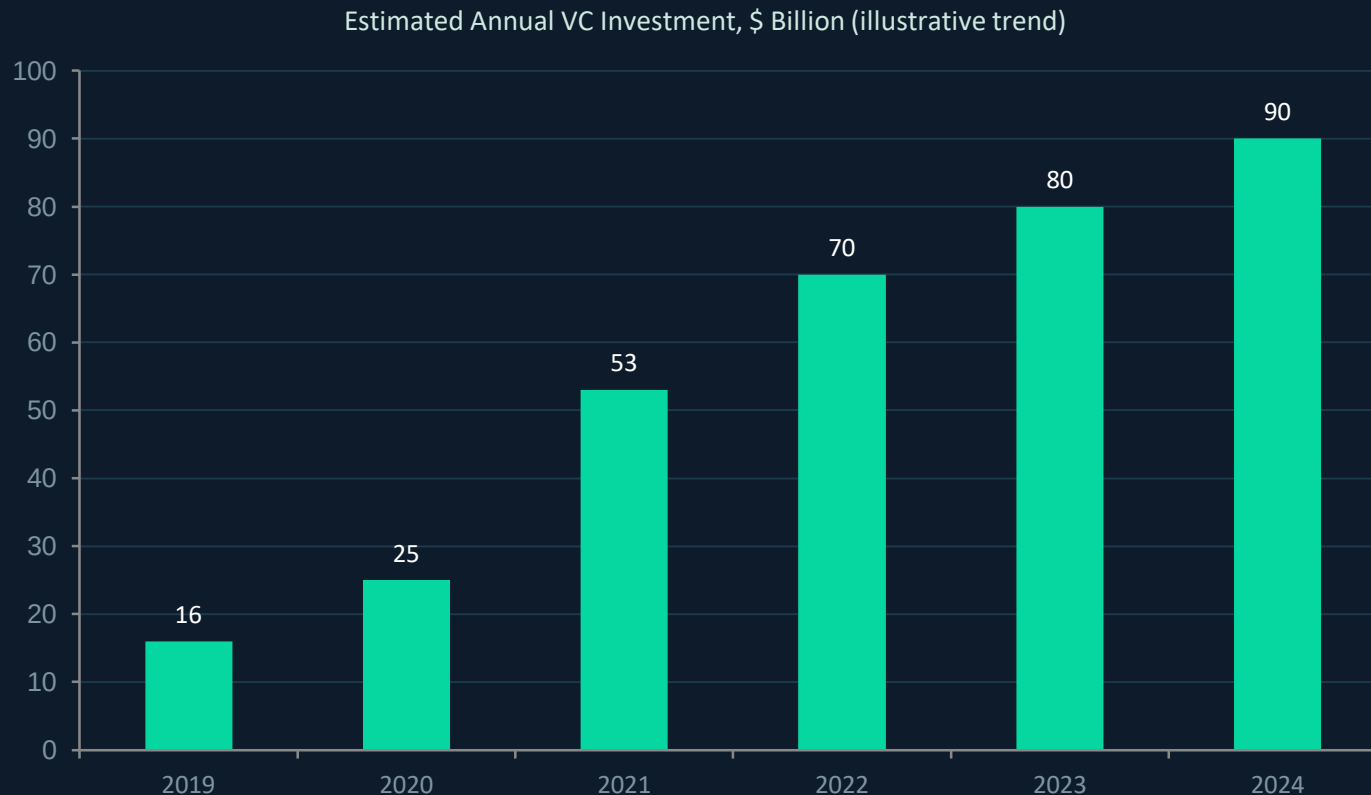
Behind every climate statistic is a computing pipeline.

Satellites capture it. Sensors stream it. Models simulate it. Software decides what happens next.

THE MARKET

A trillion-dollar computing problem

Global climate-tech investment has grown into one of the largest capital flows in technology — and it runs on engineers, not ecologists alone.



30M+

New green jobs projected globally by 2030

~40%

Of those roles require software, data or AI skills

SECTION ONE

Nine Technologies Rewriting Climate Action

AI · Data Science · IoT · Digital Twins · Robotics · Cloud · Cybersecurity · Blockchain · Software Engineering

THE PROPOSED MATRIX

Nine Technologies, Mapped to Nine Problems

Before the deep dives: the framework connecting each technology to the specific climate problem it solves, and the shift in approach it represents.

TECHNOLOGY	THE EXISTING PROBLEM	THE PROPOSED SHIFT
 AI & Machine Learning	Slow, compute-heavy prediction	Learned models — seconds, not hours
 Data Science & Analytics	Manual, annual, error-prone reporting	Automated, continuous pipelines
 Internet of Things	Sparse, infrequent physical-world data	Real-time sensor networks
 Digital Twins	Build first, discover problems later	Simulate first, build once
 Robotics & Automation	Human labor, limited scale & safety	Autonomous, continuous operation
 Cloud Computing	Supercomputer access is scarce	On-demand, elastic compute for all
 Cybersecurity	Legacy grids built with zero defense	Purpose-built OT threat monitoring
 Blockchain	Opaque, fraud-prone carbon markets	Transparent, verifiable ledgers
 Software Engineering	Closed, siloed climate datasets	Open, shared, well-engineered APIs

01



AI & Machine Learning

The predictive brain of the climate system — forecasting, optimizing, and discovering faster than any human team.

THE PROBLEM

Traditional weather and climate prediction solves physics equations on supercomputers — accurate, but slow (hours per forecast) and accessible only to a handful of national agencies.

THE PROPOSED SOLUTION

Learn patterns instead of solving equations

Train models directly on decades of historical atmospheric data so they learn the patterns physics simulations compute from scratch every time.

Forecast in seconds, not hours

A trained model can generate a 10-day global forecast in under a minute, on a single machine — no supercomputer queue required.

Extend the same approach beyond weather

The same learned-pattern approach now optimizes energy grids and predicts emissions and financial risk.

01



AI & Machine Learning

-40%

Reduction in data center cooling energy after DeepMind applied the same ML approach to Google's own cooling systems.

CASE STUDY

Google DeepMind — GraphCast

A graph neural network trained on 40 years of reanalysis weather data. It predicts 10-day global weather in under a minute, outperforming the world's leading physics-based model (ECMWF's HRES) on the majority of tested variables.

OLD MODEL vs. NEW MODEL

THE EXISTING MODEL

Physics-based Numerical Weather Prediction: hours of dedicated supercomputer time per forecast, run by a handful of national meteorological agencies worldwide.

THE PROPOSED MODEL

ML-based forecasting: under 60 seconds on a single TPU/GPU, in principle deployable by any team with access to training data.

WHY THIS SHOULD EXCITE YOU

The field of "AI for weather and climate" barely existed five years ago. A genuinely novel model architecture is still a realistic student research project today — not a solved problem you'd be repeating.

02



Data Science & Analytics

Turning billions of scattered climate signals into decisions leaders can act on.

THE PROBLEM

Corporate and national emissions reporting has long relied on annual, manually compiled, self-reported estimates — slow, inconsistent, and easy to get wrong or game.

THE PROPOSED SOLUTION

Automate the data pipeline

Continuously ingest utility bills, IoT sensor feeds, and supply-chain data instead of collecting it once a year by hand.

Calculate emissions near real time

Turn a process that once took months into a dashboard that updates automatically as new data arrives.

Flag anomalies automatically

Statistical and ML models catch reporting errors and undisclosed emissions spikes that manual review would miss.

02



Data Science & Analytics

2.5 Qt

Bytes of Earth-observation and sensor data now generated daily that platforms like this increasingly draw on.

CASE STUDY

Watershed

A carbon accounting SaaS platform used by companies including Airbnb and Block to automate what used to be a multi-month manual audit into a continuously updated emissions dashboard.

OLD MODEL vs. NEW MODEL

THE EXISTING MODEL

Annual manual spreadsheet audits, often outsourced to consultants, published 6–12 months after the reporting period already ended.

THE PROPOSED MODEL

Continuous, API-driven emissions dashboards that update automatically as new operational data arrives.

WHY THIS SHOULD EXCITE YOU

Every company on Earth will eventually need a pipeline like this. The market for it is still being built, not yet saturated — early movers are defining the category.

03



Internet of Things (IoT)

Millions of low-cost sensors turning the physical world into a real-time climate dashboard.

THE PROBLEM

Utilities and farmers have traditionally operated on sparse data — a monthly meter reading, a manual soil check — making it impossible to catch problems like leaks or drought stress until well after they start.

THE PROPOSED SOLUTION

Deploy dense, continuous sensor networks

Low-cost sensors stream data constantly instead of being checked on a calendar schedule.

Enable automated, real-time response

Systems can trigger demand-response signals or precision irrigation the moment conditions change — not weeks later.

Push decisions to the edge

Many IoT climate systems now process data locally, reacting in milliseconds without waiting on a round trip to the cloud.

03



Internet of Things (IoT)

29 Bn

IoT-connected devices projected worldwide by 2030 — a huge share feeding climate and energy systems.

CASE STUDY

CropX

A soil-sensor IoT network used by farmers worldwide to cut water use significantly while improving yields, by irrigating only where and when the underlying soil data actually calls for it.

OLD MODEL vs. NEW MODEL

THE EXISTING MODEL

Monthly manual meter readings; calendar-based (not data-based) irrigation scheduling regardless of actual soil conditions.

THE PROPOSED MODEL

Real-time sensor networks driving automated, minute-by-minute irrigation and consumption decisions.

WHY THIS SHOULD EXCITE YOU

IoT-for-climate is as much a hardware, firmware, and networking problem as a software one — there is room here for nearly every CSE specialization.

04



Digital Twins

Living, simulated replicas of cities, grids and factories — where climate strategies are tested before they cost a rupee.

THE PROBLEM

Cities and industrial facilities have historically been designed and built first — only discovering flooding, overheating, or energy-inefficiency problems afterward, when fixing them is disruptive and expensive.

THE PROPOSED SOLUTION

Build a high-fidelity virtual replica first

Model the physical system in enough detail to simulate real climate stresses against it.

Simulate before construction, not after

Test heatwaves, flooding, and energy loads virtually, catching flaws while they still cost nothing to fix.

Keep the twin live post-construction

Twins continue tracking the real asset afterward, predicting maintenance needs and failure points.

04



Digital Twins

\$13 Bn

Projected market for industrial digital twins, growing fastest in energy and infrastructure sectors.

CASE STUDY

Virtual Singapore

A detailed 3D digital twin of the entire city-state, used by planners to simulate heatwave impact, flooding, and solar potential before greenlighting real infrastructure spending.

OLD MODEL vs. NEW MODEL

THE EXISTING MODEL

Build first, retrofit later — trial-and-error in the physical world, at a cost measured in disrupted cities and wasted concrete.

THE PROPOSED MODEL

Simulate extensively in the virtual world first; only build what has already been stress-tested against climate scenarios.

WHY THIS SHOULD EXCITE YOU

Every digital twin is, at its core, a distributed-simulation and data-engineering problem — a natural fit if you enjoy systems, graphics, or large-scale modeling.

05



Robotics & Automation

Machines doing the dull, dangerous and large-scale climate work humans physically cannot.

THE PROBLEM

Reforestation and ocean cleanup have long depended on human labor — slow, expensive, and capped at a tiny fraction of the scale actually needed, versus the millions of hectares and tons involved.

THE PROPOSED SOLUTION

Automate what humans can't scale

Deploy drones, autonomous vessels, and vision-guided robots to work at industrial scale, not volunteer-crew scale.

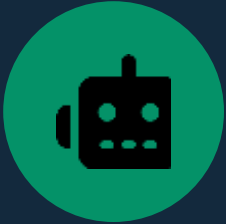
Operate in conditions humans can't sustain

Robots work continuously in heat, remote terrain, or open ocean without shift limits or safety exposure.

Run around the clock

Unlike human crews, autonomous systems don't stop for weather, daylight, or rest.

05



Robotics & Automation

24/7

Operating capability of autonomous cleanup and sorting systems — work that never stops for weather or shifts.

CASE STUDY

The Ocean Cleanup

Autonomous vessel systems that have removed millions of kilograms of plastic from the Pacific Ocean, operating continuously without needing a full-time onboard crew.

OLD MODEL vs. NEW MODEL

THE EXISTING MODEL

Small human crews, limited by weather, safety, and cost, realistically able to cover only a tiny fraction of the affected area.

THE PROPOSED MODEL

Autonomous systems operating around the clock, at a small fraction of the marginal cost per hectare or kilogram removed.

WHY THIS SHOULD EXCITE YOU

Climate robotics is still mostly unsolved outside a few flagship projects — huge open problems remain in perception, navigation, and long-term durability.

06



Cloud Computing

The planetary-scale compute layer that makes climate modeling and coordination even possible.

THE PROBLEM

Until recently, running a serious climate model required an allocation on one of a handful of national supercomputing centers — a queue of weeks to months, inaccessible to most researchers, let alone students or startups.

THE PROPOSED SOLUTION

Rent compute instead of queuing for it

Provision massive, elastic capacity by the hour from commercial cloud providers.

Run the same class of simulation, for less

Ensemble climate models that once needed a supercomputing grant now run on rented cloud HPC clusters.

Coordinate distributed assets, not just compute

Cloud platforms also coordinate thousands of rooftop solar and battery assets as a single "virtual power plant."

06



Cloud Computing

1000s

Of virtual servers spun up in minutes to run climate ensembles that once took months to schedule.

CASE STUDY

Cloud-based climate ensembles

Climate researchers and startups increasingly run large ensemble simulations on commercial cloud HPC clusters, spinning up thousands of virtual servers in minutes for workloads that once required a supercomputing allocation.

OLD MODEL vs. NEW MODEL

THE EXISTING MODEL

Apply for time on a national supercomputer; wait weeks to months for an allocation; share capacity with every other funded project.

THE PROPOSED MODEL

Provision cloud HPC capacity on demand, pay only for what is used, and scale down the moment the run finishes.

WHY THIS SHOULD EXCITE YOU

You can now run a genuinely serious climate simulation from a laptop and a credit card. The barrier to entry that once required an institutional affiliation has essentially collapsed.

07



Cybersecurity

Protecting the grids, dams and sensor networks that a decarbonized world will depend on completely.

THE PROBLEM

Power grids, water treatment plants, and renewable assets were designed assuming they'd never touch the internet — "security by obscurity." As they digitize for efficiency, that assumption has quietly become false, with almost no built-in defense.

THE PROPOSED SOLUTION

Monitor OT, not just IT

Purpose-built operational technology (OT) security watches industrial control traffic for anomalies specific to energy infrastructure.

Assume connectivity, not obscurity

Design defenses for grids that are — and will remain — network-connected, rather than pretending they are air-gapped.

Protect data integrity, not just uptime

Verifying that emissions and climate data hasn't been tampered with is now its own security discipline.

07



Cybersecurity

+150%

Rise in reported cyberattacks on energy infrastructure over the past few years as grids digitize.

CASE STUDY

Dragos

An industrial cybersecurity company focused specifically on protecting power grids, water systems, and manufacturing plants from cyberattacks, used by utilities across North America and Europe.

OLD MODEL vs. NEW MODEL

THE EXISTING MODEL

Generic IT security tooling — or often no monitoring at all — on systems assumed to be "air-gapped" and therefore unreachable.

THE PROPOSED MODEL

OT-specific threat detection purpose-built for the unique protocols and failure modes of energy infrastructure.

WHY THIS SHOULD EXCITE YOU

This is one of the most under-supplied specializations in all of cybersecurity — most security engineers today have never touched an industrial control system.

08



Blockchain

A transparent, tamper-proof ledger for a carbon economy that has long struggled with trust.

THE PROBLEM

Voluntary carbon credit markets have long been plagued by double-counting, opaque registries, and outright fraud — buyers often cannot verify that the reduction they paid for is real, or hasn't already been sold to someone else.

THE PROPOSED SOLUTION

Record credits on a shared, tamper-evident ledger

Every credit's origin and full ownership history becomes independently verifiable by anyone.

Cut out the middleman where it adds no value

Peer-to-peer platforms let households trade surplus solar power directly, without a utility intermediary.

Extend traceability up the supply chain

The same ledger approach verifies the origin of "green" materials, from cobalt to sustainable timber.

08



Blockchain

\$2 Bn+

Value flowing through blockchain-based carbon-market and renewable-trading platforms today.

CASE STUDY

Power Ledger

A blockchain-based platform enabling households with rooftop solar to trade surplus energy directly with neighbors, without a utility as middleman — deployed in Australia, Japan, Thailand, and beyond.

OLD MODEL vs. NEW MODEL

THE EXISTING MODEL

Centralized, often opaque registries prone to double-counting; utility-mediated energy trading with no peer-to-peer option.

THE PROPOSED MODEL

Transparent, cryptographically verifiable ledgers enabling direct, trustless peer-to-peer trading.

WHY THIS SHOULD EXCITE YOU

Trust infrastructure for climate markets is still being invented — this is genuinely greenfield territory, even within an otherwise mature blockchain field.

09



Software Engineering

The invisible discipline that turns every idea above from a prototype into something that actually ships and scales.

THE PROBLEM

For decades, climate and environmental datasets sat locked inside individual institutions — incompatible formats, no shared APIs — forcing every new project to rebuild the same data pipeline from scratch.

THE PROPOSED SOLUTION

Engineer for sharing, not just for use

Build well-documented public APIs instead of one-off internal data formats.

Let an ecosystem build on shared infrastructure

Thousands of independent developers can innovate on the same underlying data instead of duplicating collection effort.

Bring engineering rigor to every idea on this list

Every model, sensor network, and trading platform in this lecture is only as reliable as the software holding it together.

09



Software Engineering

1

Discipline every technology on this list depends on to go from lab demo to planetary-scale deployment: yours.

CASE STUDY

Climate TRACE

A coalition combining AI and satellite data that built an open API tracking greenhouse gas emissions from over 660 million individual sources worldwide — letting any developer build on top of it instead of gathering the data themselves.

OLD MODEL vs. NEW MODEL

THE EXISTING MODEL

Bespoke, closed data silos — every team re-solves the same integration problem before any real climate work can begin.

THE PROPOSED MODEL

Shared, well-engineered open APIs (Climate TRACE, NASA) that an entire ecosystem of builders works on top of.

WHY THIS SHOULD EXCITE YOU

This is the least glamorous and most essential technology on the list — and the one almost every other climate-tech job actually needs on a daily basis.

THE BREADTH, AT A GLANCE

More builders putting this into practice



Tesla Autobidder

AI software trades and dispatches battery power across grids in real time.



Climeworks

Software-orchestrated direct air capture plants pulling CO₂ from the atmosphere.



Google DeepMind

AI cut Google data center cooling energy use by up to 40%.



Watershed

A data platform automating enterprise carbon accounting at scale.



ChargePoint

IoT-connected EV charging network coordinating millions of sessions.



Pachama

Satellite imagery + AI verify forest-carbon offset projects globally.

SECTION TWO

Your Future in Climate Technology

Research frontiers, startup opportunities, and the careers this decade is creating for engineers like you.

Four doors into climate technology



Research

M.Tech / PhD in climate informatics, geospatial ML, energy systems, and environmental data science at IITs, IISc, MIT, Stanford.



Startups

Founding or joining early-stage climate-tech ventures — carbon accounting, grid software, agri-tech, climate risk analytics.



Corporate Climate Tech

Dedicated sustainability engineering teams at Google, Microsoft, Tesla, Schneider Electric, and every major energy utility.



Policy & Tech Advisory

Technical roles in government, WEF, UNFCCC and think tanks that need engineers who can build, not just advise.

**The climate doesn't need
more environmentalists.
It needs more engineers like you.**

Every technology in this lecture was built by someone who once sat exactly where you are sitting now.

UNIT 1 · TOPIC 1

Introduction to Climate Change

The story of how we discovered we were changing our own planet — and why it starts with you

COURSE

In your grandparents' lifetime, humans changed the chemistry of the entire atmosphere.

Not a country. Not a continent. The entire planet's air — measurably, permanently, in about 70 years.

That is the story of this lecture.

Six Stops on the Way to Understanding



01

Why This Moment Matters



02

What Do We Mean by "Climate Change"?



03

How Do We Know? — The Evidence



04

A Short History of a Long Discovery



05

What's Driving It? — A First Look



06

Why This Should Matter to You

Numbers That Should Stop You for a Second

+1.1°C

Global average warming since the late 1800s — most of it in the last 40 years

420 ppm

Atmospheric CO₂ today — last seen roughly 3–4 million years ago

1.2 cm

Sea level is rising per year on average, and accelerating

8

Of the 10 hottest years ever recorded have occurred since 2015

None of these numbers are projections. They are measurements — already on record.

Source: NASA GISS; NOAA; IPCC AR6 (2021–2023 synthesis).

"Has anyone in this room noticed a change in weather, seasons, or climate within their own lifetime — even something small?"

Turn to the person next to you. One example each. Be ready to share.

🕒 3 MIN

A Planet That Has Never Changed This Fast

Earth's climate has always changed — but the speed of the current change is what makes it unprecedented.

Natural Change Is Slow

Past ice ages and warm periods unfolded over thousands to tens of thousands of years, giving ecosystems time to adapt or migrate.

This Change Is Fast

The current pace of warming is estimated to be at least 10 times faster than the fastest natural warming after the last ice age.

Why Speed Matters

Evolution, migration, and human adaptation all take time. A planet that changes in decades, not millennia, leaves little room to adjust.

SECTION 2 OF 6

What Do We Mean by "Climate Change"?

A term used constantly in the news — and defined precisely, and specifically, in science.

What Is Climate Change, Exactly?

The IPCC and the UNFCCC define it slightly differently — and the difference matters.

IPCC Definition

Any change in the state of the climate, identifiable by changes in its mean and/or variability, persisting for decades or longer — regardless of cause.

UNFCCC Definition

A change in climate attributed directly or indirectly to human activity, altering atmospheric composition, in addition to natural variability.

Why It Matters

The UNFCCC's narrower, human-caused definition is the one used in international climate policy, law, and negotiation — including everything in Units ahead.

Weather vs. Climate — In Brief



Weather

What the atmosphere is doing right now, in one place.

Changes hour to hour, day to day.

"It is unusually hot in Delhi today."



Climate

The long-term pattern and average of weather, over decades.

Changes gradually, over decades to centuries.

"Delhi's climate is getting hotter, decade over decade."

Myth or Fact?

MYTH

"Climate change" and "global warming" mean exactly the same thing.

Global warming is one symptom (rising temperature). Climate change is the broader phenomenon — also including shifting rainfall, extreme weather, and sea level rise.

MYTH

A single freezing week in Delhi disproves global warming.

One region's short-term weather event says nothing about a decades-long global average trend — this is one of the most common reasoning errors in the public debate.

MYTH

Scientists are still genuinely divided on whether this is happening.

Over 97% of actively publishing climate scientists agree recent warming is real and predominantly human-caused — the debate in the literature is about impact and response, not existence.

FACT

Earth's climate has changed naturally before, many times.

True — ice ages and warm periods are part of Earth's history. What is different now is the rate of change and the clearly identified human cause.

SECTION 3 OF 6

How Do We Know? — The Evidence

Not belief. Not opinion. Independent lines of physical evidence that all point the same direction.



Six Ways We Measure a Changing Climate

Different tools, different teams, different decades — and they all tell the same story.



Direct Thermometers

Global surface temperature records dating back to the 1880s, from thousands of stations worldwide.



Ice Cores

Ancient air bubbles trapped in Antarctic and Greenland ice reveal atmospheric CO₂ going back 800,000 years.



Satellites

Since 1979, satellites have independently measured temperature, sea level, and ice extent from orbit.



Glacier Retreat

Photographed and measured glacier boundaries worldwide show consistent, accelerating retreat.



Sea Level Gauges

Tide gauges and satellite altimetry both confirm accelerating sea level rise since the early 20th century.

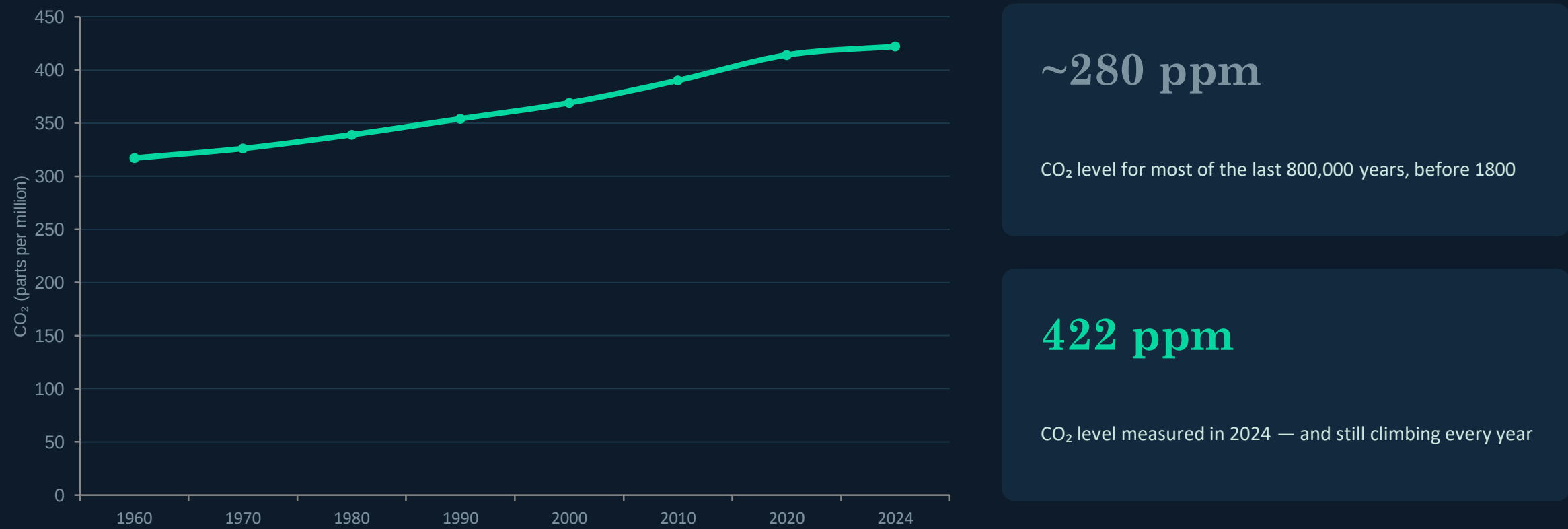


Tree Rings & Corals

Natural growth records extend the climate record thousands of years before instruments existed.

The Keeling Curve

Continuous atmospheric CO₂ measurement at Mauna Loa Observatory, Hawaii — running since 1958 without interruption.



Named for Charles David Keeling, who began this measurement in 1958. Data: Scripps CO₂ Program & NOAA (illustrative).

"Of the six independent evidence types we just saw, which one do you personally find most convincing — and why?"

There is no wrong answer. The point is that six unrelated fields all reached the same conclusion.

🕒 2 MIN

Ice cores allow scientists to measure atmospheric CO₂ going back how far in time?

A About 1,000 years

B About 50,000 years

C About 800,000 years



D About 10 million years

ANSWER & WHY

C. About 800,000 years

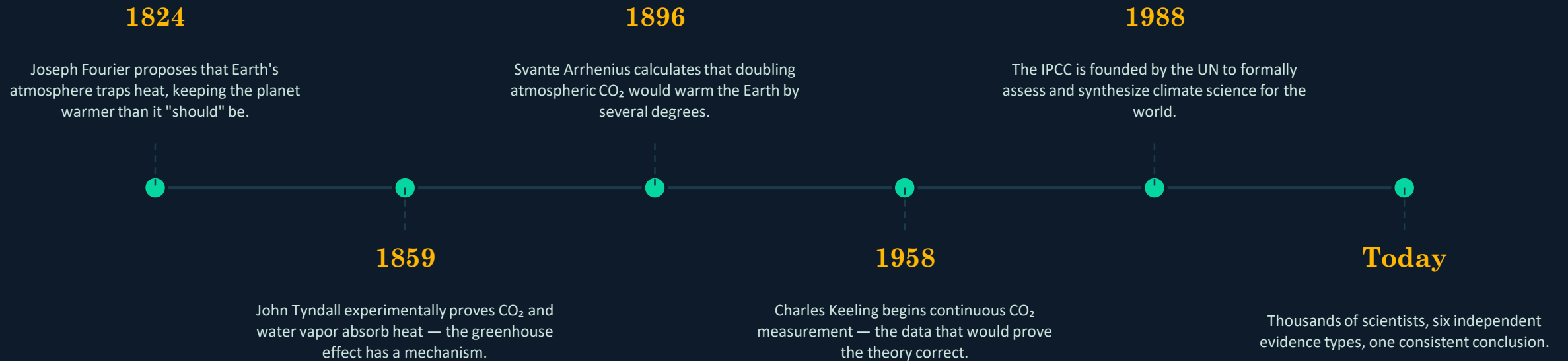
The longest continuous Antarctic ice cores (such as EPICA Dome C) preserve trapped air bubbles reaching back roughly 800,000 years — giving us a direct physical sample of the ancient atmosphere itself, not a model or estimate.

SECTION 4 OF 6

A Short History of a Long Discovery

This was not discovered last year. Scientists have been building this case for 200 years.

Two Centuries in the Making



The Scientists Behind the Discovery

Climate science was not built by one person, or one country, or one era.

Joseph Fourier (France, 1824)

A mathematician and physicist who first asked why Earth is warmer than simple physics predicts.

Eunice Foote & John Tyndall (US/UK, 1856–59)

Foote first showed CO₂ traps heat; Tyndall independently proved it rigorously in the lab a few years later.

Svante Arrhenius (Sweden, 1896)

A chemist who, entirely by hand, calculated the warming effect of CO₂ — a calculation later confirmed by supercomputers.

Which scientist first proposed, in 1824, that Earth's atmosphere traps heat?

A Svante Arrhenius

B Joseph Fourier



C Charles Keeling

D John Tyndall

ANSWER & WHY

B. Joseph Fourier

Joseph Fourier, better known to engineers for the Fourier Transform, was also the first to reason — using basic physics — that Earth's atmosphere must be trapping heat for the planet to be as warm as it is.

SECTION 5 OF 6

What's Driving It? — A First Look

A preview only — the full mechanics come later in this unit. For now, the headline picture.



Natural Causes vs. Human Causes



Natural Factors

Solar cycles, volcanic eruptions, and orbital variations all influence climate.

These operate on their own natural cycles, mostly over centuries to millennia.

Estimated net effect since 1950: roughly neutral to slightly cooling.



Human (Anthropogenic) Factors

Burning fossil fuels, deforestation, and industrial agriculture release greenhouse gases.

These have surged dramatically since the Industrial Revolution — especially post-1950.

Estimated net effect since 1950: the dominant driver of observed warming.

Natural or Human-Driven?

Classify each factor — is it a natural driver of climate, or a human-driven one?

- 1 Burning coal for electricity generation
- 2 Variations in the sun's energy output over an 11-year cycle
- 3 Large-scale deforestation for agriculture
- 4 A major volcanic eruption releasing ash and aerosols
- 5 Vehicle and aviation emissions
- 6 Earth's slow, cyclical changes in orbit (Milankovitch cycles)

NATURAL

2

4

6

HUMAN-DRIVEN

1

3

5

SECTION 6 OF 6

Why This Should Matter to You

The motivational core of today's lecture — because understanding is only step one.



This Is Not a Distant, Future Problem

3.3–3.6 Bn

People already live in contexts highly vulnerable to climate impacts

21.5 M

People displaced annually, on average, by weather-related events since 2008

1 in 4

Days per year the average person now experiences extreme heat, up sharply since the 1980s

This generation — yours — will make more consequential decisions about this than any before it.

Source: IPCC AR6 (2022); UNHCR; World Weather Attribution (illustrative synthesis).

Climate Change Is Now a Computer Science Problem

Every major climate solution today runs, in some way, through code.

Climate Models

Today's climate projections run on some of the world's largest supercomputing clusters — pure computational science.

Data at Planetary Scale

Satellites, sensors, and monitoring stations generate more climate data every day than humanity can currently process.

AI-Accelerated Solutions

From weather prediction to grid optimization to carbon accounting — machine learning is now central to climate response.

The Opportunity

You are entering this field at the exact moment it needs computer scientists more than it ever has before.

"If you spent your career working on exactly one climate-related problem, which part of it — data, modeling, infrastructure, policy tech — would you want to own?"

No wrong answers. Just start noticing where your own curiosity points.

🕒 3 MIN

What We Covered Today

1

The Scale

Climate change is already measurable — not a future projection alone.

2

The Definition

A long-term shift in climate state, with the UNFCCC definition specifically pointing to human causes.

3

Weather vs. Climate

The moment versus the long-run average and pattern.

4

The Evidence

Six independent methods — thermometers to tree rings — all agree.

5

The Keeling Curve

The single most important continuous dataset in climate science.

6

The History

Two centuries of accumulating scientific evidence, not a recent claim.

7

The Drivers

Natural cycles exist, but human activity is the dominant force since 1950.

8

The Stakes

Billions of people already live with real, current climate exposure.

9

Your Role

This is now inseparable from computing, data, and AI — your field.

10

Next Step

Bring your curiosity — Topic 2 goes deeper into Weather vs. Climate and Earth's Climate System.

Questions & Discussion

"What is one thing from today's lecture you want to know more about — and why?"

Next: Weather vs. Climate & Earth's Climate System

Thank you

Sustainability and Climate Change · Unit 1, Topic 1 · B.Tech Final Year, CSE

B.TECH FINAL YEAR · COMPUTER SCIENCE & ENGINEERING · OPEN ELECTIVE

UNIT 1 · TOPICS 2–5

The Science Engine of Climate Change

Earth's Climate System · The Greenhouse Effect · Global Warming · The Carbon Cycle

COURSE

Sustainability & Climate Change

SESSION FOCUS

The Mechanics Behind the Headlines

FORMAT

Lecture + In-Class Activities

BEFORE WE BEGIN

Four mechanisms. One planet. Every climate headline you've ever read traces back to these.

A system. A physical mechanism. A measured trend. A planetary-scale feedback loop — this is climate change as an engineer sees it: not a belief, but a chain of cause and effect you can actually trace.

BY THE END OF THIS SESSION

You will be able to...



Model

Describe Earth's climate system as five interacting spheres in dynamic energy balance.



Explain

Derive why the greenhouse effect is necessary for life — and how it becomes dangerous when amplified.



Interpret

Read real temperature and CO₂ datasets, and distinguish natural from human-driven warming.



Trace

Follow carbon through Earth's fast and slow cycles, and pinpoint exactly where humans broke the balance.

Four Systems, Four Stories



01

Earth's Climate System

The five spheres that make climate a system, not a single number.



02

The Greenhouse Effect

The natural process that keeps us alive — and the one we've amplified.



03

Global Warming

The measured, accelerating trend — and what is really driving it.



04

The Carbon Cycle

The planet's carbon ledger, and how humans overdrew the account.

SECTION 1 OF 4

Earth's Climate System

Five interacting spheres, one planetary engine — and the energy balance that drives it all.



The Components of Earth's Climate System

Climate emerges from constant interaction between five great subsystems of the planet.



Atmosphere

The blanket of gases surrounding Earth — where weather happens and greenhouse gases trap heat.



Hydrosphere

All of Earth's water — oceans, lakes, rivers, and groundwater — which stores and transports vast amounts of heat.



Cryosphere

Frozen water — ice sheets, glaciers, sea ice, and permafrost — which reflects sunlight and locks away carbon.



Lithosphere

Earth's solid crust and upper mantle, including soil and rock, which slowly cycles carbon over geological time.

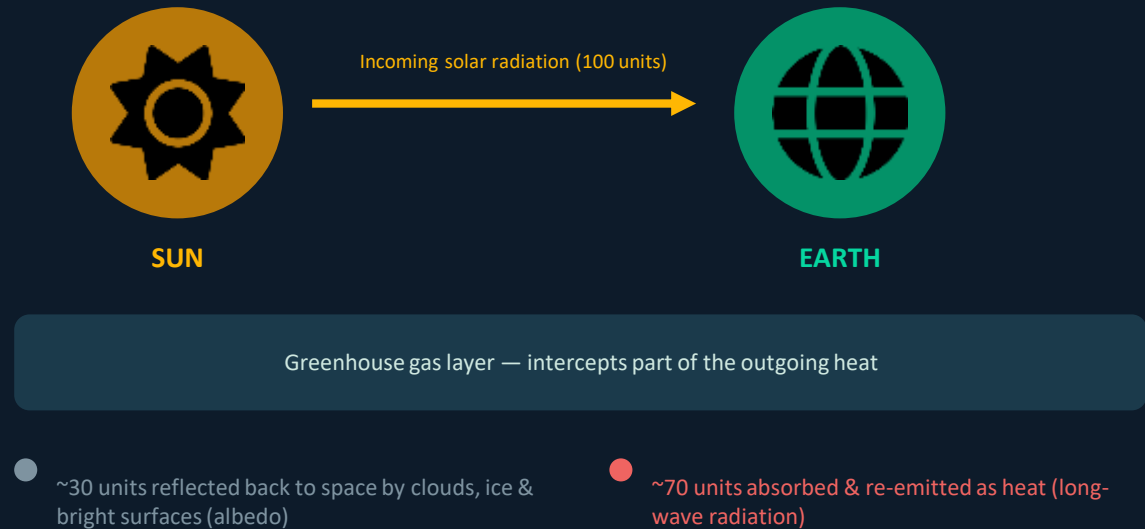


Biosphere

All living organisms — plants, animals, and microbes — that absorb and release carbon through life processes.

Earth's Energy Balance

For Earth's temperature to stay stable, incoming solar energy must roughly equal outgoing energy. Climate change disrupts this balance.



WHY THE BALANCE MATTERS

- Roughly 30% of incoming solar energy is reflected straight back to space by clouds, ice, and bright surfaces (this is called albedo).
- The remaining ~70% is absorbed by Earth and re-radiated as heat (long-wave/infrared radiation).
- Greenhouse gases absorb some of that outgoing heat and re-emit it back down — this is the natural greenhouse effect.
- More greenhouse gases means more heat trapped, and the balance point shifts to a warmer Earth.

Climate as Earth's Largest Distributed System

As computer scientists, you already have the mental model for this — you just haven't applied it here yet.

Nodes & State

The five spheres are like distributed nodes, each holding and exchanging state (heat, carbon, water) with the others.

Feedback Loops

Melting ice reduces reflectivity, which increases absorbed heat, which melts more ice — a positive feedback loop, exactly like a runaway process in a control system.

Latency & Buffering

The ocean acts as a massive buffer, absorbing and slowly releasing heat over years to decades — much like a cache with very high latency.

Why It's Hard

Like any large distributed system, small local changes can propagate into large global effects — making the system genuinely difficult to predict precisely.

SECTION 2 OF 4

The Greenhouse Effect

The natural process that makes life possible — and the mechanism behind global warming when amplified.



The Greenhouse Effect Keeps Us Alive

Without any greenhouse effect at all, Earth would be a frozen, uninhabitable rock.

Without It

Earth's average temperature would be roughly -18°C — about 33°C colder than it is today. Oceans would freeze.

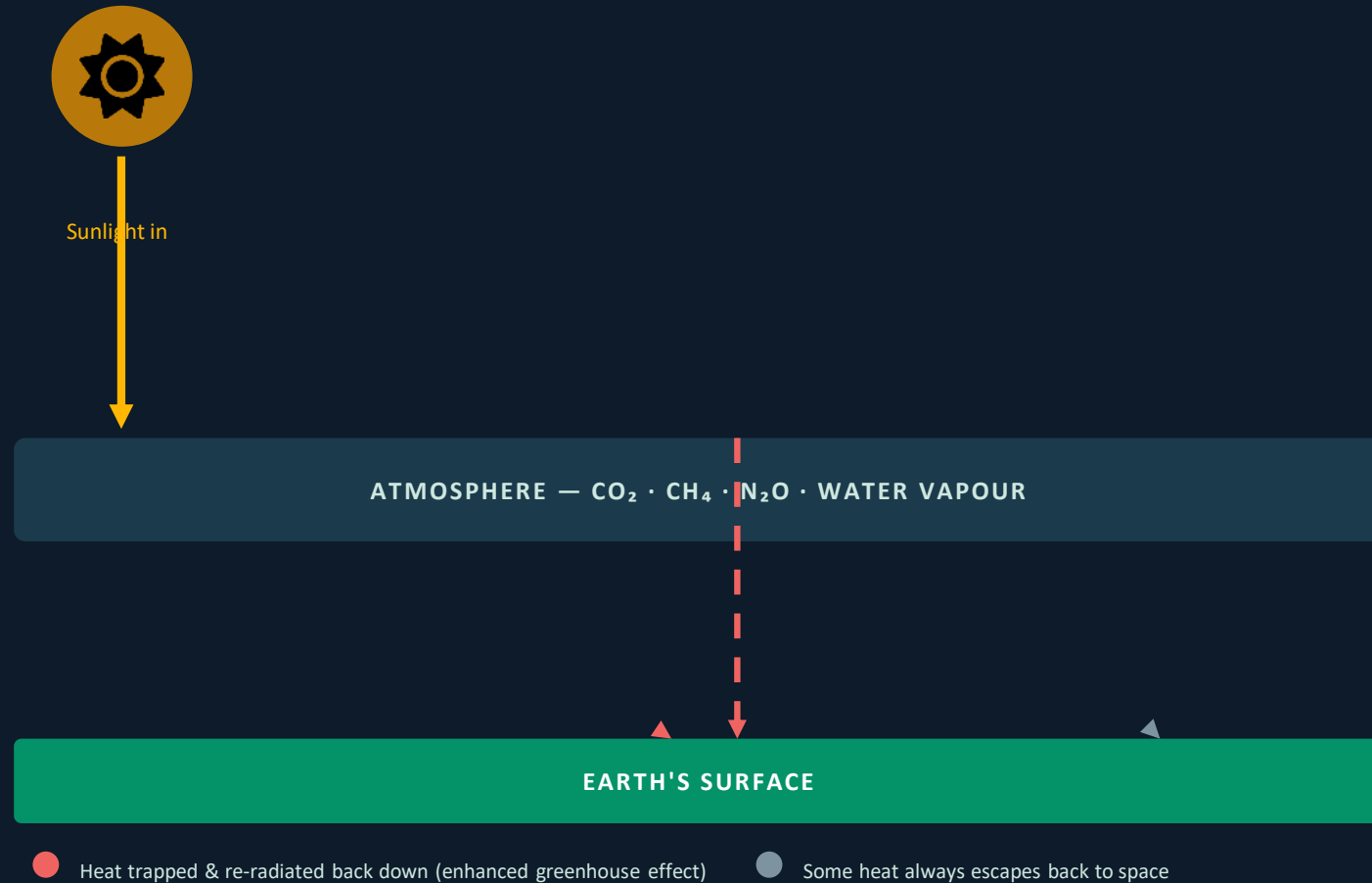
With the Natural Effect

Greenhouse gases trap just enough heat to keep Earth's average temperature around a life-friendly $+15^{\circ}\text{C}$.

The Problem Isn't the Effect Itself

The problem is that human emissions have intensified this natural process well beyond its historical equilibrium — an enhanced, human-amplified greenhouse effect.

How the Greenhouse Effect Works



THE FOUR-STEP PROCESS

1. Sunlight arrives

Short-wave solar radiation passes through the atmosphere largely unimpeded.

2. Earth absorbs & re-emits

The surface absorbs this energy and re-radiates it as long-wave infrared (heat).

3. Greenhouse gases intercept it

CO₂, methane, and other gases absorb this outgoing infrared radiation.

4. Heat is re-radiated back

Gases re-emit that energy in all directions — some back to the surface — warming the planet.

Major Greenhouse Gases & Their Sources

Not all greenhouse gases are equal — they differ hugely in abundance, lifetime, and heat-trapping strength.

Carbon Dioxide (CO₂)

Fossil fuel combustion, deforestation, cement production. ~76% of human GHG emissions by volume — the primary driver due to sheer quantity.

Methane (CH₄)

Livestock digestion, rice paddies, landfills, natural gas leaks. Traps far more heat per molecule than CO₂, but breaks down faster (~12 years).

Nitrous Oxide (N₂O)

Synthetic fertilizers, industrial processes. Roughly 265–298 times more warming per molecule than CO₂ over a 100-year period.

Water Vapour & F-gases

Water vapour is the largest natural greenhouse gas but responds to warming rather than driving it; F-gases (from refrigerants, industry) are synthetic and extremely potent.

Global Warming Potential (100-Year, vs. CO₂ = 1)

1

Carbon Dioxide (CO₂) — the reference gas

27–30

Methane (CH₄) — short-lived, but very potent

273

Nitrous Oxide (N₂O) — from fertilizers & industry

25,200

Sulfur Hexafluoride (SF₆) — used in electrical insulation

CO₂ dominates warming only because of volume, not potency — gram for gram, it is the weakest of these four.

Source: IPCC AR6 Working Group I (2021), 100-year Global Warming Potential values.

Myth or Fact?

MYTH

The greenhouse effect itself is a bad thing that we need to eliminate.

The natural greenhouse effect is essential for life. The problem is the human-caused enhancement of it — not its existence.

FACT

Methane is a more potent greenhouse gas than CO₂, molecule for molecule.

Over 100 years, methane traps roughly 27–30 times more heat per molecule than CO₂ — though it breaks down much faster.

MYTH

CO₂ is the most dangerous greenhouse gas because it traps the most heat per molecule.

CO₂ is actually one of the weaker gases per molecule — its danger comes from the sheer volume humans emit, and its long atmospheric lifetime.

FACT

Water vapour is technically the largest contributor to the natural greenhouse effect.

Water vapour traps more heat than any other greenhouse gas — but it responds to temperature (a feedback), rather than independently driving warming like CO₂.

Which greenhouse gas has the highest Global Warming Potential (GWP) per molecule over 100 years, among common gases?

A Carbon Dioxide (CO₂)

B Methane (CH₄)

C Nitrous Oxide (N₂O)

D Sulfur Hexafluoride (SF₆)



ANSWER & WHY

D. Sulfur Hexafluoride (SF₆)

SF₆, used in electrical insulation, has a GWP of about 25,200 — by far the most potent common greenhouse gas. CO₂ is the reference gas (GWP = 1) and dominates only because of its enormous volume, not its per-molecule potency.

SECTION 3 OF 4

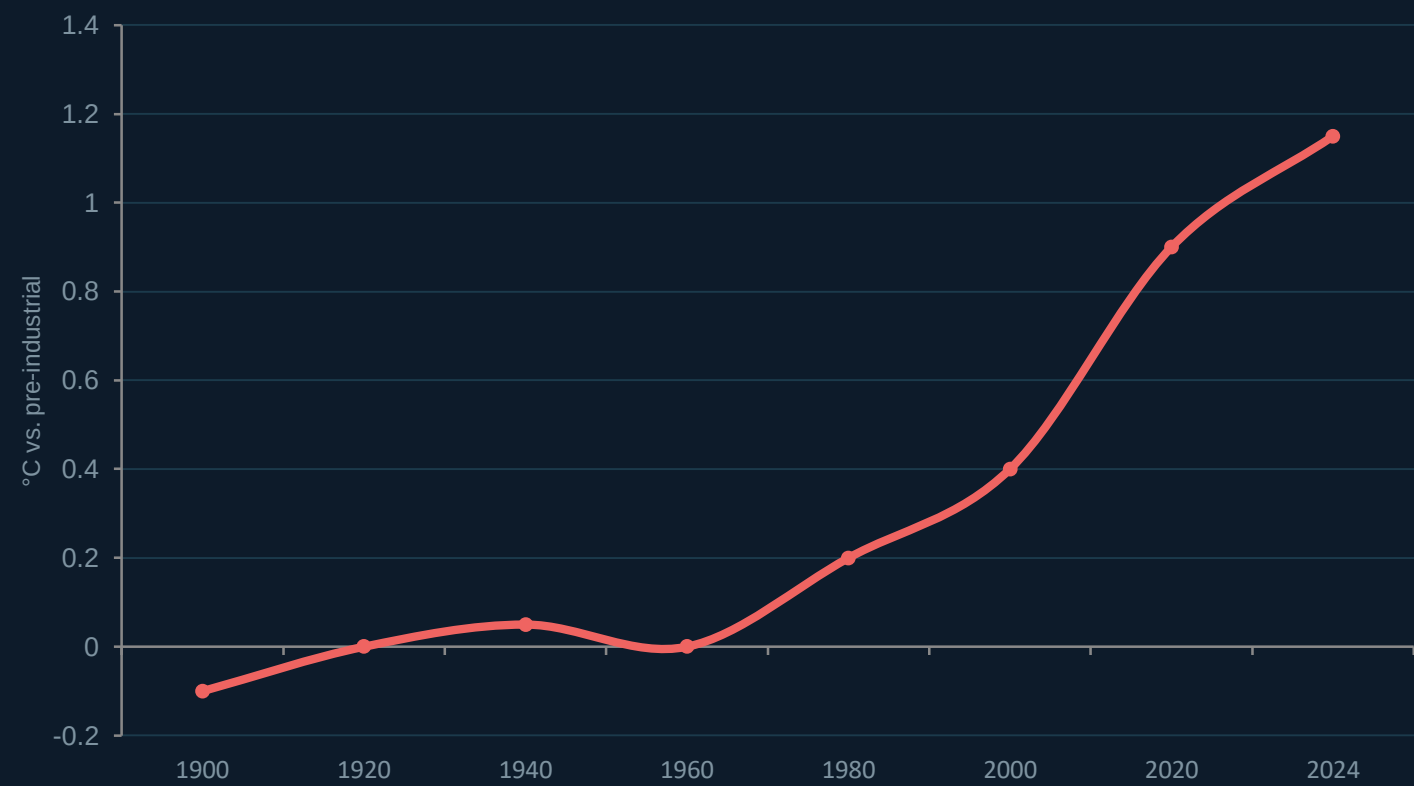
Global Warming

The measurable, accelerating rise in global average temperature — and what is driving it.



Global Temperature Is Rising — And Accelerating

Global average surface temperature relative to the 1850–1900 pre-industrial baseline.



+1.1°C

Current warming vs. pre-industrial levels

2011–2020

The warmest decade on record

Illustrative trend based on IPCC AR6 and NASA GISS surface temperature datasets.

What Is Driving Global Warming?



Natural Factors

Solar cycles — minor fluctuations in the sun's output over ~11-year cycles.

Volcanic eruptions — can cause short-term cooling via reflective particles.

Earth's orbital variations (Milankovitch cycles) — operate over tens of thousands of years.

Net effect since 1950: roughly neutral to slightly cooling.



Human (Anthropogenic) Factors

Burning fossil fuels (coal, oil, gas) for energy, industry, and transport.

Deforestation — removes carbon sinks and releases stored carbon.

Industrial agriculture — livestock, fertilizer use, and rice cultivation.

Net effect since 1950: the dominant driver — responsible for virtually all observed warming.

Feedback Loops & Tipping Points

This is where your background in systems and nonlinear dynamics becomes directly useful.

Positive Feedback: Ice-Albedo

Less ice → less sunlight reflected → more heat absorbed → more ice melts. A self-reinforcing loop, just like unstable positive feedback in a control system.

Positive Feedback: Permafrost Carbon

Warming thaws permafrost → releases trapped methane and CO₂ → causes more warming → thaws more permafrost.

Tipping Points

Beyond certain thresholds, these feedback loops can become self-sustaining even if human emissions stopped — the climate equivalent of a system that has left its stable operating region.

Why This Matters for Prediction

Nonlinear systems with feedback loops are notoriously hard to forecast precisely — which is exactly why climate models require immense computational power.

"Can you think of a feedback loop from your own field — software, networks, or otherwise — that behaves like ice-albedo or permafrost carbon?"

Runaway cache invalidation? Cascading failures? The pattern is more familiar than you think.

📄 3 MIN

Looking at the temperature trend chart, which statement best describes the pattern since 1980?

- A Temperature has risen at a slow, constant rate since 1900
- B Temperature has fluctuated randomly with no clear trend
- C The rate of warming has clearly accelerated in the last four decades ✓**
- D Temperatures peaked in 1980 and have been falling since

ANSWER & WHY

C. The rate of warming has clearly accelerated in the last four decades

The chart shows a relatively flat line through the mid-20th century, followed by a steepening climb from around 1980 onward — consistent with the rapid rise in global greenhouse gas emissions during that period.

SECTION 4 OF 4

The Carbon Cycle

Carbon moves constantly between air, land, ocean and life — until human activity tipped the balance.



The Carbon Cycle

Carbon continuously flows between four great reservoirs. Under natural conditions, what leaves one enters another in balance.



Atmosphere

~875 GtC



Land / Biosphere

~2,000 GtC

KEY FLUXES BETWEEN RESERVOIRS

- **Photosynthesis**

Plants absorb atmospheric CO₂ to build biomass (Atmosphere → Biosphere).

- **Respiration & Decay**

Organisms release CO₂ back to the atmosphere as they breathe and decompose.

- **Ocean Exchange**

Oceans absorb and release CO₂ at the surface; the ocean is currently a net carbon sink.

- **Combustion (Human)**

Burning fossil fuels releases carbon locked away for millions of years — the cycle's biggest disruption.



Ocean

~38,000 GtC



Fossil Fuels & Rock

~10,000,000 GtC

The Fast Cycle vs. the Slow Cycle



Fast Carbon Cycle

Moves carbon between atmosphere, ocean surface, and living things.

Timescale: days to centuries.

Example: a tree absorbing CO₂ through photosynthesis.

This is the cycle humans have dramatically disrupted.



Slow Carbon Cycle

Moves carbon between rocks, sediments, and the deep ocean.

Timescale: thousands to millions of years.

Example: carbon slowly locked into limestone or fossil fuel deposits.

Fossil fuel combustion pulls slow-cycle carbon into the fast cycle — the core of the problem.

How Humans Have Unbalanced the Carbon Cycle

The carbon cycle balanced itself for millennia. Two human activities changed that.

Fossil Fuel Combustion

Releases carbon that took millions of years to accumulate underground, on a timescale of decades — far faster than natural sinks can reabsorb it.

Deforestation & Land-Use Change

Destroys biosphere carbon sinks and releases stored carbon, while also reducing Earth's future capacity to absorb CO₂.

Net Result

Atmospheric CO₂ has risen from ~280 ppm (pre-industrial) to over 420 ppm today — the highest level in at least 800,000 years.

The Global Carbon Budget Is a Ledger Problem

Think of it as an accounting system running a structural deficit.

The Budget Concept

Scientists estimate a finite "carbon budget" — the total CO₂ humanity can still emit while keeping warming under 1.5°C or 2°C.

Withdrawals vs. Deposits

Fossil fuel combustion is a withdrawal; reforestation and carbon capture are deposits. Right now, withdrawals vastly outpace deposits.

Why This Resonates With CS

It is fundamentally a resource-accounting and optimization problem — tracking flows, sources, and sinks at planetary scale, which is exactly the kind of system engineers are trained to model and manage.

Sort the Step: Source or Sink?

For each process below, is it currently releasing carbon into the atmosphere (a source) or removing it (a sink)?

		SOURCE	SINK
1	Burning coal in a power plant		
2	Photosynthesis in a tropical rainforest	1	2
3	Ocean absorbing atmospheric CO ₂	4	3
4	Large-scale deforestation for agriculture		
5	Soil organic carbon storage in grasslands		
6	Cement production (calcination of limestone)	6	5

Which of these best describes why fossil fuel combustion is so disruptive to the carbon cycle?

- A It creates a new type of carbon molecule never seen before
- B It moves slow-cycle carbon into the fast cycle far faster than natural processes can absorb it ✓**
- C It only affects the atmosphere, not the ocean or land
- D It has no measurable long-term effect on atmospheric CO₂

ANSWER & WHY

B. It moves slow-cycle carbon into the fast cycle far faster than natural processes can absorb it

Fossil fuels represent carbon that was locked away in the slow cycle over millions of years. Burning them in the span of decades injects that carbon into the fast cycle far quicker than photosynthesis, ocean uptake, and rock weathering can remove it again.

The Four Mechanisms, Connected

1

Five Spheres

Earth's climate system is atmosphere, hydrosphere, cryosphere, lithosphere, and biosphere in constant interaction.

2

Energy Balance

Incoming solar energy (~100 units) versus outgoing heat (~70 absorbed/re-emitted, ~30 reflected) sets the planet's temperature.

3

A Necessary Effect

The greenhouse effect keeps Earth ~33°C warmer than it would otherwise be — essential, not inherently harmful.

4

Human Amplification

CO₂, methane, N₂O, and F-gases have intensified this natural process well past historical equilibrium.

5

Measured Warming

+1.1°C since pre-industrial times, accelerating sharply since 1980 — not a projection, a measurement.

6

Feedback Loops

Ice-albedo and permafrost carbon feedbacks can self-reinforce warming — the same nonlinear dynamics you study in systems.

7

Fast vs. Slow Carbon

Nature separates carbon into fast (days–centuries) and slow (millennia+) cycles — and fossil fuels bridge the two dangerously.

8

Human Disruption

Fossil combustion and deforestation moved carbon from the slow cycle into the fast cycle faster than sinks can absorb it.

9

A Ledger Problem

The global carbon budget is a resource-accounting system currently running a structural deficit.

10

Your Toolkit Applies

Systems thinking, feedback loops, and data modeling — skills you already have — are exactly what this problem needs.

**The planet runs on systems, feedback loops,
and data — the exact tools you are learning.
You are more qualified for this than you think.**

Every mechanism in today's lecture — energy balance, feedback loops, carbon accounting — is a systems problem wearing a climate-science costume.

From Concern to Action

Global Concerns · Sectoral Impacts · Responses & Mitigation · Climate Protocols & Agreements

**You now understand the mechanism.
Today: the stakes, and who is actually doing something
about it.**

Governments, companies, treaties, and — increasingly — engineers like you. This is where climate science turns into climate action.

From Global Stakes to Global Rules



01

Global Concerns

The planet-scale risks now shaping migration, markets, and policy.



02

Impacts Across Sectors

No sector is untouched — including the one you are about to join.



03

Responses & Mitigation

Two strategies, many actors — from national governments to your own choices.



04

Climate Protocols & Agreements

The evolving legal architecture behind global climate action.

SECTION 1 OF 4

Global Concerns

From melting ice caps to shifting monsoons — the tangible, planet-wide stakes.



Key Global Concerns

Climate change is not one problem — it is a cluster of interconnected, planet-wide risks.



Sea Level Rise

Melting ice sheets and thermal expansion threaten coastal cities and small island nations.



Extreme Weather

More frequent and intense heatwaves, floods, droughts, and storms.



Biodiversity Loss

Shifting habitats and temperatures push species toward extinction faster than they can adapt.



Food & Water Security

Changing rainfall and heat stress threaten crop yields and freshwater availability.



Climate Migration

Millions may be displaced by sea level rise, drought, and resource conflict in coming decades.



Public Health

Heat stress, disease vector spread, and air quality decline all worsen with warming.



Air Quality

Many drivers of climate change also directly worsen air pollution and respiratory illness.



Wildfires

Hotter, drier conditions extend fire seasons and increase their frequency and intensity.

The Scale of Global Concern

3.3–3.6 Bn

People live in contexts highly vulnerable to climate impacts (IPCC AR6)

21.5 M

People displaced annually on average by weather-related events since 2008

~50%

Of the world population faces severe water scarcity at least one month per year

1 Million+

Species at risk of extinction, with climate change a major compounding driver

These are not distant, future risks — they are already shaping migration, markets, and policy today.

Source: IPCC AR6 (2022); UNHCR; UN Water; IPBES Global Assessment (illustrative synthesis).

Every Global Concern Is Also a Data Problem

Before any of these risks can be managed, they first have to be measured, modeled, and monitored — at planetary scale.

Early Warning Systems

Flood, heatwave, and cyclone early-warning systems are fundamentally real-time data pipelines — sensors, models, and alerts racing against time.

Satellite Monitoring

Deforestation, glacier retreat, and drought are tracked largely through satellite imagery pipelines processed with computer vision and ML.

Migration & Risk Modeling

Predicting climate-driven displacement requires modeling human behavior against environmental data — a hard, unsolved computational problem.

The Opportunity

Every concern on the previous slide has an unfilled software and data engineering gap behind it.

"Which of these global concerns do you think will affect your hometown the most in the next 20 years — and why?"

Discuss with the person next to you, then be ready to share one insight with the class.

🕒 3 MIN

According to IPCC AR6, roughly how many people already live in contexts highly vulnerable to climate impacts?

A About 500 million

B About 1 billion

C Between 3.3 and 3.6 billion



D Nearly the entire global population

ANSWER & WHY

C. Between 3.3 and 3.6 billion

The IPCC's Sixth Assessment Report estimates that 3.3 to 3.6 billion people — roughly 40–45% of the world's population — already live in contexts highly vulnerable to climate change, concentrated in Africa, South Asia, and Central/South America.

SECTION 2 OF 4

Impacts Across Sectors

No industry is untouched — including the one most of you are about to join.



Climate Impacts Across Sectors

Every major sector of the economy faces distinct, material climate risk.



Agriculture

Shifting rainfall, heat stress, and pest ranges threaten yields and farmer livelihoods worldwide.



Water Resources

Glacial melt and erratic monsoons strain freshwater supply for drinking, farming, and industry.



Public Health

Heat-related illness, vector-borne disease, and air pollution impose rising healthcare burdens.



Infrastructure

Roads, power grids, and buildings face stress from floods, heat, and extreme weather events.



Energy

Both energy demand (cooling) and supply (hydro, thermal plant cooling water) are climate-sensitive.



Tourism & Aviation

Coastal and snow tourism destinations face existential risk; aviation faces both impact and mitigation pressure.



Insurance & Finance

Rising climate losses are repricing risk across insurance, banking, and real estate markets.



Digital Infrastructure

Data centers need constant cooling and power — both increasingly climate-constrained resources.

Climate Change Meets Digital Infrastructure

This is the sector you are entering. It is not exempt — it is deeply exposed.

Data Center Cooling

Data centers already consume ~1–1.5% of global electricity, with cooling as a major share. Rising ambient temperatures increase cooling costs and strain reliability.

Water Stress

Many hyperscale data centers use water-based cooling. Regions facing water scarcity are increasingly restricting or taxing data center water use.

Grid Reliability

Extreme weather increasingly threatens power grid stability — exactly the infrastructure cloud services depend on for uptime.

The Flip Side

The same sector causing part of the problem is also building the tools — AI, IoT, digital twins — used to solve it. That tension is now central to the profession you are entering.

Which Sector Does This Impact Describe?

Match each described impact to the sector it primarily affects.

1	Erratic monsoon timing disrupts sowing and harvest cycles	AGRICULTURE	INFRASTRUCTURE	HEALTH
2	Urban flooding damages roads, bridges, and underground cabling	1	2	3
3	Longer, more intense heatwaves increase heatstroke cases	6	5	4
4	Rising temperatures expand the range of disease-carrying mosquitoes			
5	Coastal power substations face increased flood risk			
6	Shifting rainfall patterns reduce staple crop yields			

Roughly what share of global electricity consumption do data centers currently account for?

A About 0.1%

B About 1–1.5% 

C About 10%

D About 25%

ANSWER & WHY

B. About 1–1.5%

Data centers currently consume an estimated 1–1.5% of global electricity — a figure projected to grow significantly as AI workloads expand, making energy and cooling efficiency a first-order concern for the industry you are entering.

SECTION 3 OF 4

Responses & Mitigation

Two strategies, many actors — from national governments to individual choices.



Mitigation vs. Adaptation



Mitigation

Reducing or preventing greenhouse gas emissions at the source.

Example: switching a power grid from coal to solar and wind.

Example: improving energy efficiency in buildings and industry.

Goal: address the root cause — slow or stop future warming.

Needed because: some warming is still avoidable.



Adaptation

Adjusting systems and behavior to cope with climate impacts already happening.

Example: building sea walls or flood-resistant infrastructure.

Example: developing drought-resistant crop varieties.

Goal: manage the consequences — reduce harm from current and locked-in warming.

Needed because: some warming is already locked in, regardless of mitigation.

How Governments & International Bodies Respond

National and international institutions set the frameworks within which everyone else operates.

International Bodies

The UNFCCC coordinates global negotiations; the IPCC provides the scientific assessments that inform them.

National Governments

Set emissions targets (NDCs), carbon pricing policies, renewable energy mandates, and building codes.

India's Response

The National Action Plan on Climate Change (NAPCC) coordinates 8 national missions spanning solar energy, energy efficiency, water, and sustainable habitats.

Sub-National Action

States, cities, and municipalities increasingly set their own targets — often faster-moving than national policy.

How Industry & Civil Society Respond

Governments do not act alone — markets, institutions, and individuals all shape outcomes.

Private Sector

Corporations set net-zero targets, invest in renewable energy, and face growing pressure from investors on climate risk (ESG).

NGOs & Civil Society

Organizations like WWF, Greenpeace, and thousands of local groups drive advocacy, research, and on-ground conservation.

Academia & Research

Universities and research institutions generate the science, models, and technologies mitigation and adaptation depend on.

Individuals & Communities

Consumption choices, voting behavior, and grassroots organizing shape both markets and policy from the bottom up.

Climate Tech: Where Mitigation Meets Code

A fast-growing, talent-starved segment of the tech industry — and most CS graduates aren't looking at it yet.

Grid & Energy Software

AI-driven platforms like Tesla Autobidder trade and dispatch battery power across grids in real time.

Carbon Accounting Platforms

Startups like Watershed and Persefoni use data pipelines to automate enterprise emissions reporting — a booming SaaS category.

Climate Risk Analytics

Financial institutions increasingly need software to model and price physical and transition climate risk.

The Career Signal

Climate-tech hiring has grown faster than general tech hiring in recent years, while remaining comparatively under-supplied with engineering talent.

Mitigation or Adaptation?

For each real-world action, decide whether it primarily reduces emissions (mitigation) or manages climate impacts (adaptation).

		MITIGATION	ADAPTATION
1	A city builds an early-warning flood system		
2	A country shifts its power grid to solar and wind	2	1
3	Farmers switch to drought-tolerant crop varieties	4	3
4	A company improves its factory's energy efficiency	6	
5	A coastal town relocates critical infrastructure inland		
6	A government introduces a carbon tax on emissions		5

"If a climate-tech startup offered you an internship tomorrow, which part of the problem — grid software, carbon accounting, risk modeling — would you want to work on?"

There is no wrong answer. Just notice where your curiosity goes.

📄 2 MIN

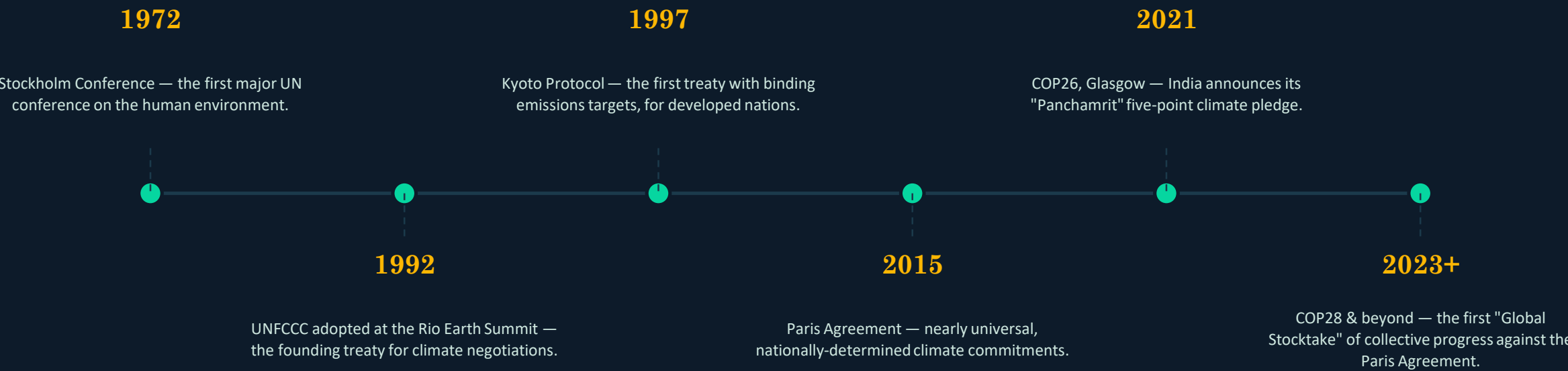
SECTION 4 OF 4

Climate Protocols & Agreements

The evolving legal and diplomatic architecture behind global climate action.



Major Climate Agreements, 1972–Present



The Paris Agreement — Key Features

Adopted in 2015 by 196 parties, the Paris Agreement remains the central framework for global climate action.

The Core Goal

Limit global warming to well below 2°C above pre-industrial levels, pursuing efforts to limit it to 1.5°C.

Nationally Determined Contributions (NDCs)

Each country sets its own emissions targets, reviewed and strengthened every five years.

Global Stocktake

A periodic review of collective progress toward the Agreement's goals, informing the next round of NDCs.

Climate Finance

Developed nations committed to mobilizing climate finance to support developing nations' mitigation and adaptation.

India's Climate Pledges (Panchamrit, COP26)

500 GW

Non-fossil energy capacity target by 2030

50%

Of energy needs from renewables by 2030

**1 Bn
Tonnes**

Projected reduction in total carbon emissions by 2030

Net Zero

India's target year for net-zero emissions: 2070

India's pledges balance ambitious climate action with its continuing development priorities.

Source: India's statement at COP26, Glasgow (2021); Ministry of Environment, Forest and Climate Change.

Why Every Protocol Needs an MRV System

A climate agreement is only as credible as the data behind it — and that data has to come from somewhere.

The MRV Problem

Measurement, Reporting, and Verification (MRV) is how the world checks whether countries and companies are actually meeting their climate pledges.

Satellite Verification

Independent satellite monitoring (e.g. methane-tracking missions) increasingly cross-checks self-reported national and corporate emissions data.

Blockchain & Carbon Markets

Emerging carbon-credit registries use distributed ledgers to reduce double-counting and fraud in voluntary carbon markets.

The Opportunity

Every treaty in this timeline ultimately depends on trustworthy software systems to prove it is actually working — that dependency is only growing.

Which climate agreement first introduced legally binding emissions targets — but only for developed nations?

A The Stockholm Declaration (1972)

B The UNFCCC (1992)

C The Kyoto Protocol (1997) ✓

D The Paris Agreement (2015)

ANSWER & WHY

C. The Kyoto Protocol (1997)

The Kyoto Protocol was the first treaty to set binding emissions reduction targets, but only for industrialized ("Annex I") countries. The Paris Agreement later moved to a more universal, nationally-determined model covering nearly all countries.

From Concern to Commitment

1

Global Concerns

Sea level rise, extreme weather, biodiversity loss, food & water security, and migration — already unfolding, not future risks.

2

A Data Problem Too

Every global concern depends on monitoring, early-warning, and modeling systems to be managed at all.

3

Sectoral Impacts

From agriculture to digital infrastructure — no sector, including yours, is untouched.

4

Your Sector, Exposed

Data centers face real cooling, water, and grid-reliability risk from the very crisis their tools help solve.

5

Mitigation vs. Adaptation

Reducing emissions at the source, versus managing the impacts already locked in — both are necessary.

6

State & Non-State Actors

Governments set frameworks; companies, NGOs, academia, and individuals all shape outcomes within them.

7

Climate Tech Careers

Grid software, carbon accounting, and climate risk analytics are fast-growing, talent-starved fields.

8

The Protocol Timeline

From Stockholm (1972) through Kyoto (1997) to Paris (2015) — an evolving architecture for collective action.

9

India's Panchamrit

500 GW non-fossil capacity, 50% renewables, and net-zero by 2070 — ambition balanced against development needs.

10

MRV Needs Engineers

Every treaty depends on trustworthy data systems to prove commitments are actually being met.